

Solutions to selected exercises from §1.4

Question 6

Fix $a, b, c \in \mathbb{R}$. So solve the problem we must find $\lambda_1, \lambda_2, \lambda_3$ such that

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}. \quad (1)$$

In other words $a = \lambda_1 + \lambda_2, b = \lambda_1 + \lambda_3, c = \lambda_2 + \lambda_3$. So

$$a - b + c = 2\lambda_2, a + b - c = 2\lambda_1, -a + b + c = 2\lambda_3. \quad (2)$$

This shows that $\lambda_1, \lambda_2, \lambda_3$ exist and hence the three vectors generated $\mathbb{R}^3$.

Question 11

A vector $v \in \text{span}\{x\}$ is in the span of the singleton set if and only if $v = \lambda x$ for some $\lambda \in \mathbb{F}$. Thus

$$\text{span}\{x\} = \{\lambda x \mid \lambda \in \mathbb{F}\}. \quad (3)$$

This is a line.

Question 12

First, suppose that $W \subset V$ is a subspace. We will show that $\text{span}(W) = W$. For any set S it is always true that $S \subset \text{span}(S)$, so to prove the assertion we must only show that $\text{span}(W) \subset W$. For this, start with $v \in \text{span}(W)$. By definition of span, there exists $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ and $u_1, \dots, u_m \in W$ such that

$$v = \lambda_1 u_1 + \dots + \lambda_m u_m. \quad (4)$$

On the other hand, since W is a subspace we know that it is closed under arbitrary addition and scalar multiplication. Thus, we see that $v \in W$ as desired.

In the other direction, assume that $W \subset V$ is a subset such that $\text{span}(W) = W$. Since we have shown that $\text{span}(S)$ is a subspace for any S , the result follows.

Question 16

In class on 02/03.